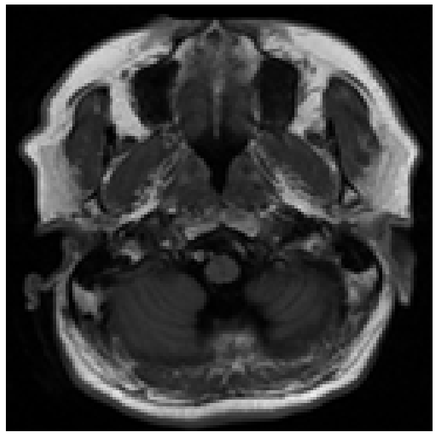
Med-CycleGAN: Frequency-Aware CT ↔ MRI Synthesis



(a) Input MRI



(b) Ground Truth CT



(c) Generated CT

EE655: Course Project

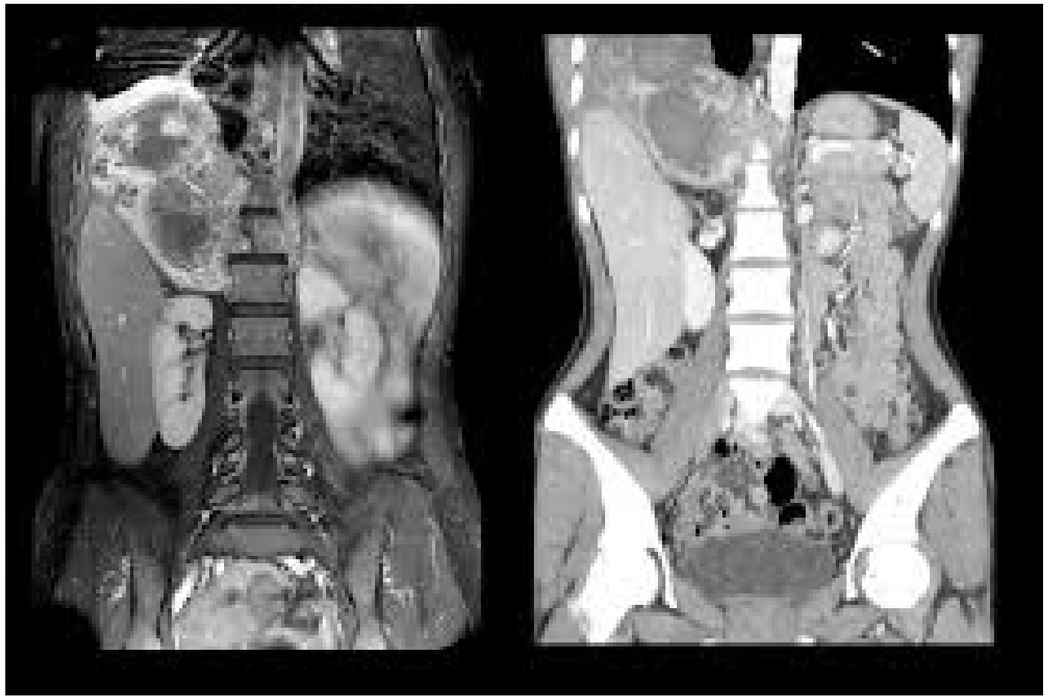
Presented By: Group 24

- Piyush Singal
- Pulkit
- Harsh Gour
- Aditya Singh
- Nithin DH

ABOUT OUR

PROBLEM STATEMENT

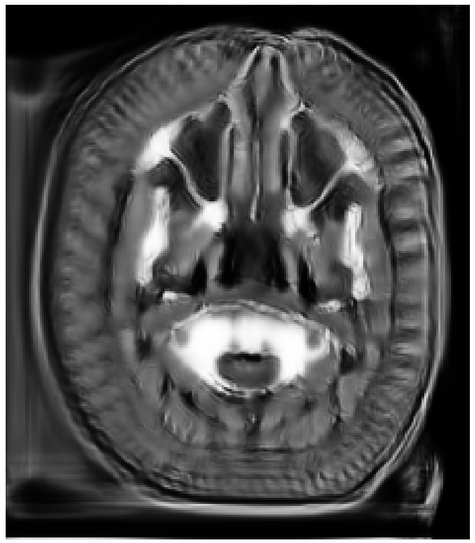
- In medical practice, both CT and MRI scans are often required
- CT → good for bones
- MRI → good for soft tissues
- Problems:
    a. Expensive
    b. Time-consuming
    c. CT exposes patients to radiation
- **Goal**: Generate one modality from the other using AI



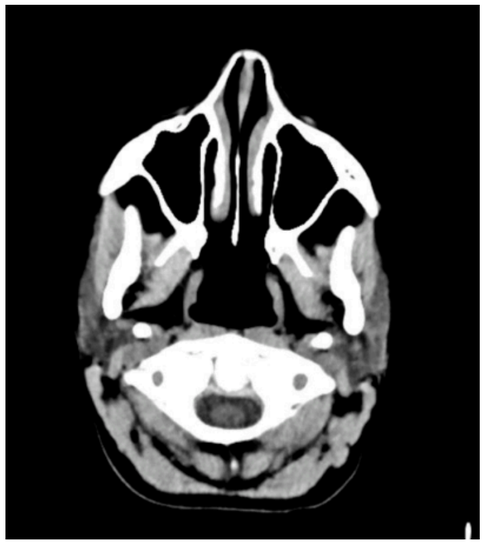
MRI Scan

EXISTING APPROACH & LIMITATION

- CycleGAN enables unpaired image translation
- Learns mapping without paired CT–MRI data
- Widely used in medical imaging
- Limitation:
  - Treats all pixels equally
  - Fails to preserve high-frequency details (edges, bones)
  - Produces blurry and clinically unreliable outputs



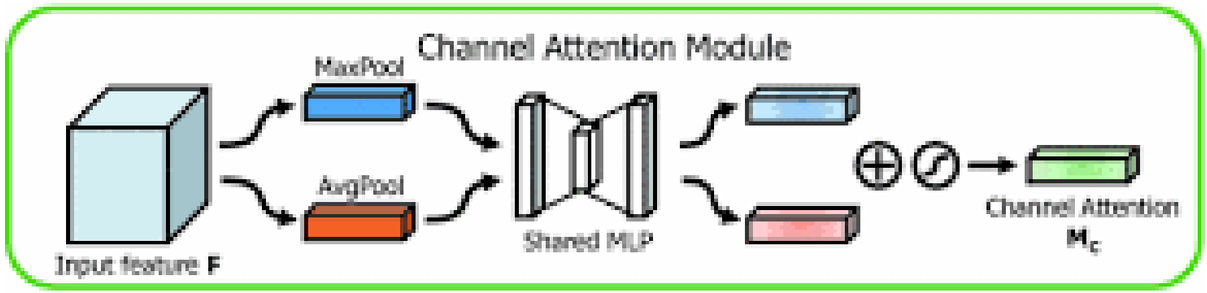
Output CT Scan

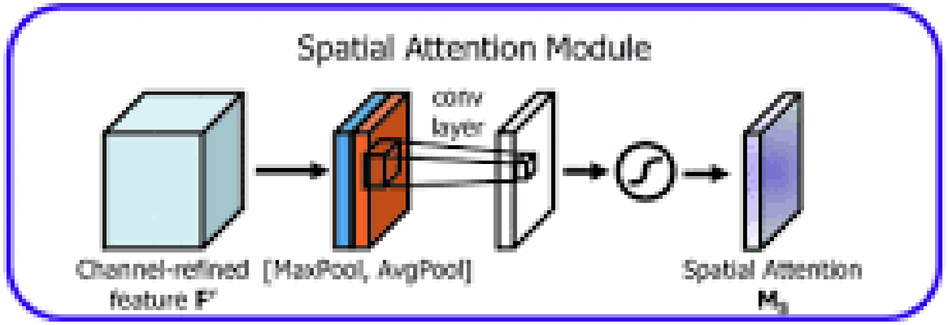


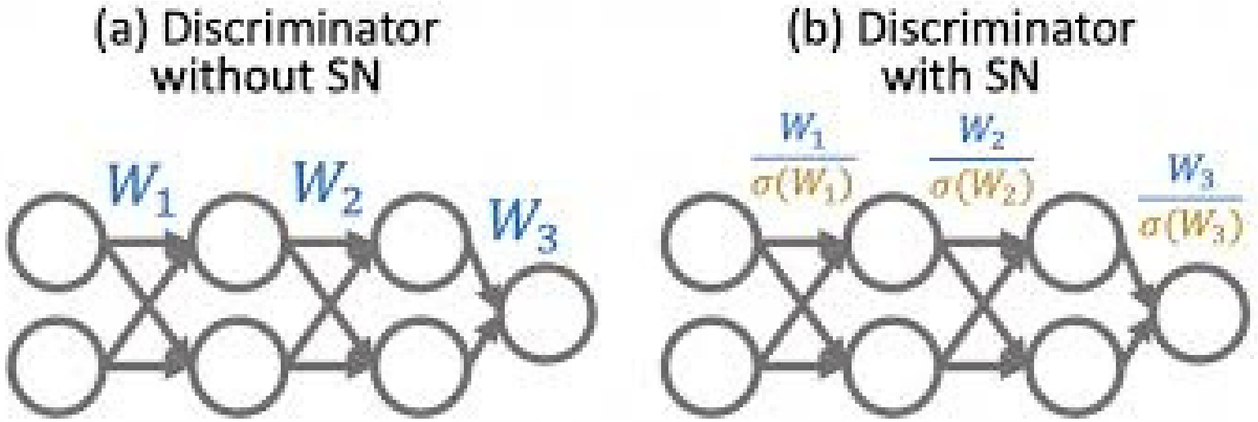
Expected CT Scan

PROPOSED METHOD

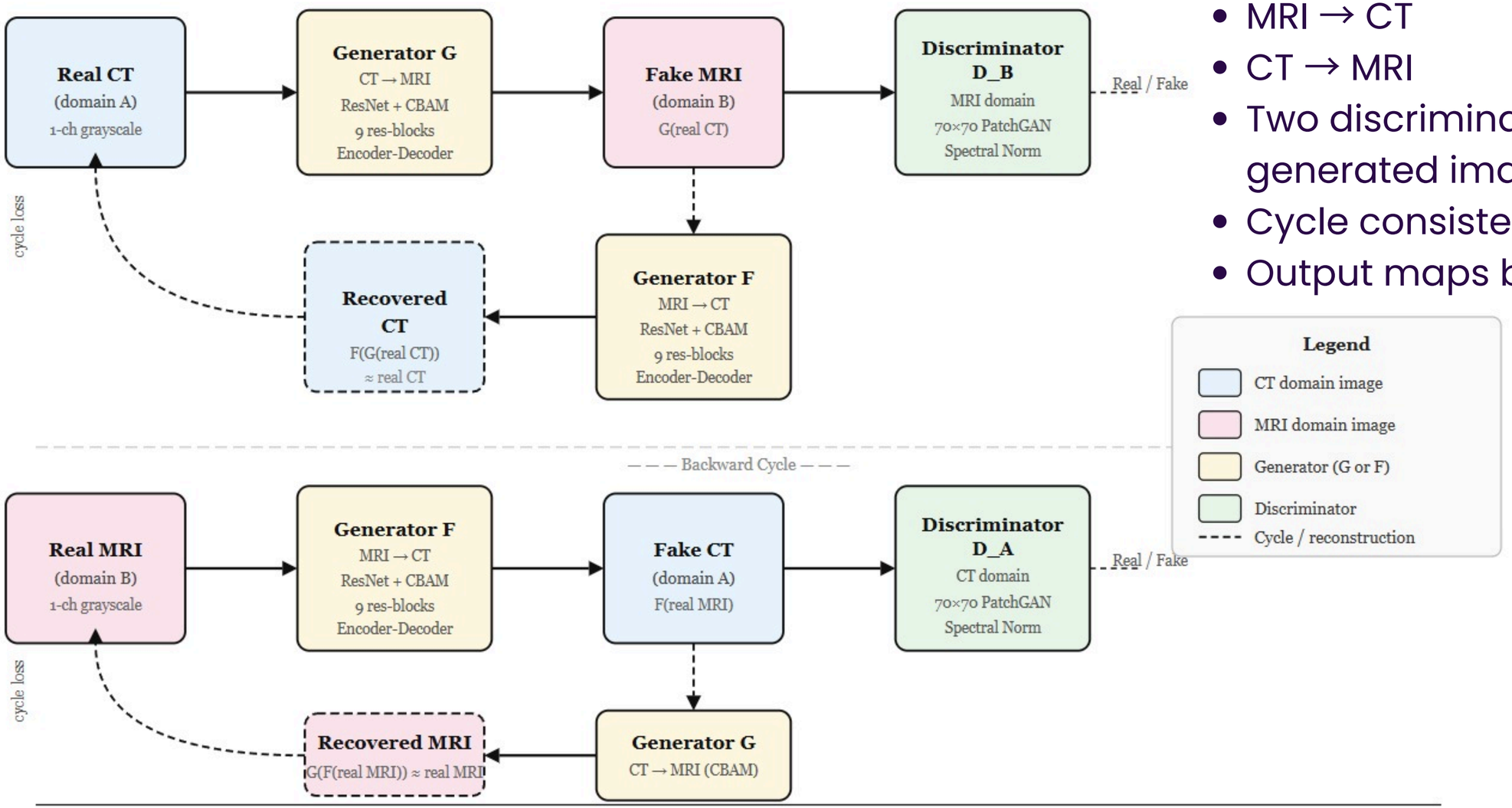
- We propose Med-CycleGAN
- Enhancements over standard CycleGAN:
  - Frequency-Domain Loss (FDL)
  - CBAM Attention modules
  - Spectral Normalization
  - Multi-objective loss for edge preservation.







ARCHITECTURE OVERVIEW



- Two generators:
- MRI → CT
- CT → MRI
- Two discriminators classify real vs generated images
- Cycle consistency ensures:
- Output maps back to original input

GENERATOR & DISCRIMINATOR

GENERATOR (CBAM-ENHANCED)

- Built on a ResNet-based architecture with integrated attention modules
- Channel Attention: Learns which feature maps (e.g., edges, textures) are important
- Spatial Attention: Learns where to focus in the image (e.g., brain regions, bone boundaries)
- Helps the model prioritize clinically relevant anatomical structures

DISCRIMINATOR (SPECTRAL NORMALIZATION)

- Uses a PatchGAN-based structure to evaluate image realism locally
- Spectral Normalization constrains weight updates, preventing unstable training
- Ensures a balanced competition between generator and discriminator

RESULT:

- Generator produces sharper, more focused images
- Training becomes stable, avoiding common GAN issues like mode collapse

LOSS FUNCTIONS

Loss Components:

- L1 Loss: Ensures pixel-wise similarity between generated and real images
- SSIM (Structural Similarity): Preserves perceptual structure (contrast, luminance)
- Gradient Difference Loss (GDL): Penalizes differences in image gradients → improves edge sharpness

Frequency Domain Loss (FDL) – Our Key Contribution:

$$F(u,v) = \sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

$$\mathcal{L}_{FDL} = \|M \odot (\mathcal{F}_{fake} - \mathcal{F}_{real})\|_1$$

- Converts images into frequency domain using 2D FFT
- Separates information into:
  - Low-frequency components: Smooth regions (soft tissues)
  - High-frequency components: Sharp edges (bones, boundaries)
- Applies higher penalty to high-frequency errors (important for clinical accuracy)

Why This Matters:

- Standard losses treat all pixels equally → leads to blurred edges
- FDL explicitly forces the model to preserve fine anatomical details
- Results in sharper bone boundaries and clearer brain structures

$$L_{cyc} = \lambda_{cyc}\left(\alpha \cdot L_{SSIM} + (1-\alpha)\cdot L_{L1} + \frac{\lambda_{edge}}{\lambda_{cyc}} L_{GDL} + \frac{\lambda_{fft}}{\lambda_{cyc}} L_{FFT}\right)$$

DATASET & METRICS

Dataset Setup:

- Used two brain imaging datasets:
  - One unpaired dataset → used for model training
  - One paired dataset → used only for evaluation
- Ensures realistic training while enabling quantitative compari

$$PSNR = 10 \cdot \log_{10}\left(\frac{MAX_I^2}{MSE}\right)$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

$$SSIM(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

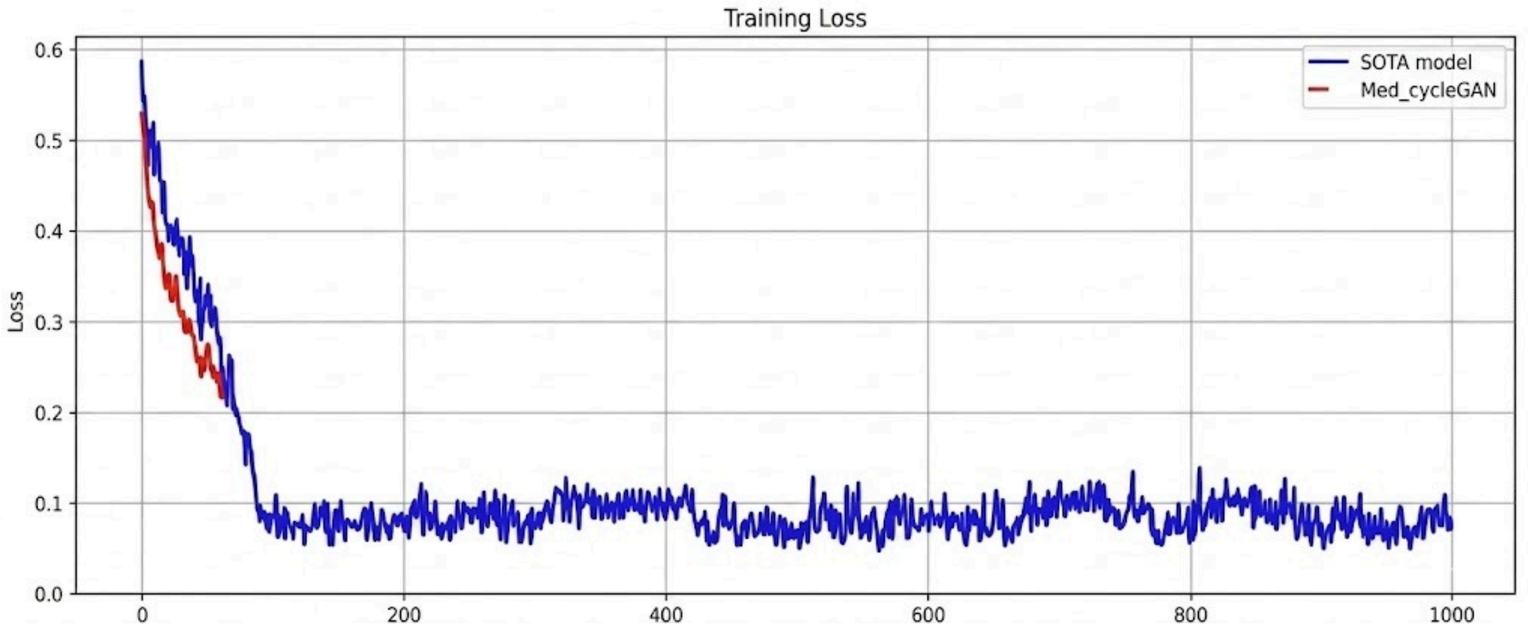
Evaluation Metrics:

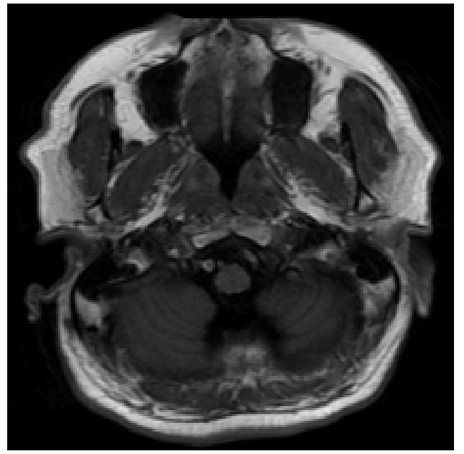
- PSNR: Measures reconstruction quality (higher is better)
- SSIM: Captures structural similarity and perceptual quality
- MAE: Measures pixel-wise error (lower is better)
- FID (Fréchet Inception Distance): How well the entire population of generated images matches the entire population of real images.

Method	PSNR ↑	SSIM ↑	FID ↓	MAE ↓
CycleGAN (Baseline)	27.4	0.731	85.2	0.058
Med-CycleGAN (Ours)	**29.5**	**0.781**	**62.4**	**0.041**
State-of-the-Art Reference (High Compute Required):				
Diffusion Models	29.7–33.1	0.910–0.947	45.0–55.0	0.017–0.025

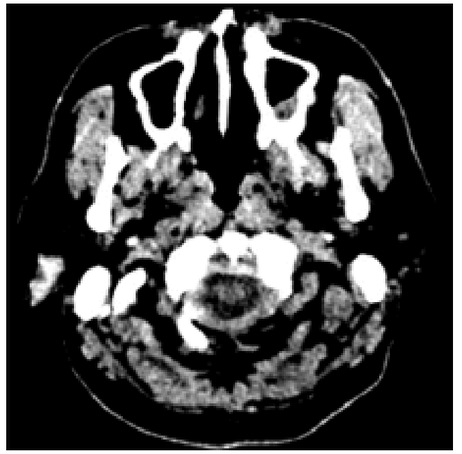
RESULTS

- Quantitative Improvements:
  - Higher PSNR and SSIM
  - Lower MAE than CycleGAN
- Qualitative Improvements:
  - Sharper anatomical boundaries
  - Better brain structure preservation

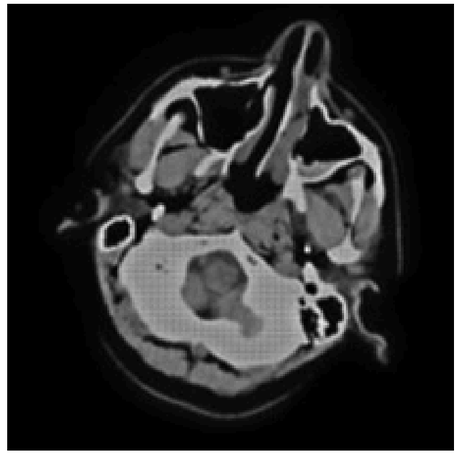




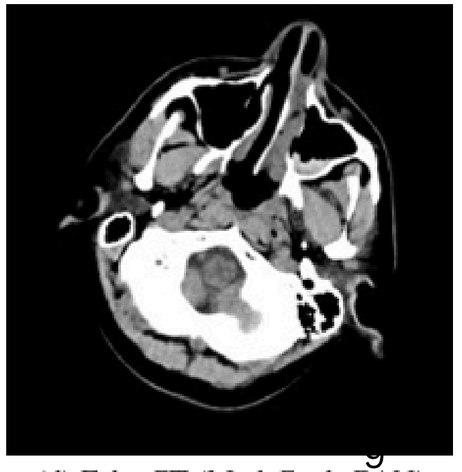
(a) Real MRI (Paired Ref.)



(b) Real CT (Paired Ref.)



(c) Real MRI (Unpaired Input)



(d) Fake CT (Med-CycleGAN)

COMPARISON & IMPACT

Compared with diffusion-based models:

Slightly lower performance

BUT much lower computational cost

Advantages:

Works on basic GPUs

Faster training

Clinically practical

CONCLUSION & FUTURE WORK

Conclusion:

Successfully improved CT ↔ MRI synthesis

Solved blurring using frequency-aware learning

Produced clinically meaningful outputs

Future Work:

Extend to 3D volumetric data

Improve realism further

Clinical validation and deployment

THANK YOU

Group 24